

Pedagogical notes on removing finite-size winding loops in exact diagonalization of quantum spin ice

July 17, 2026

Abstract

Exact diagonalization (ED) is the natural numerical route to the π -flux quantum spin ice (QSI) regime, where QMC has a sign problem. The obstacle is not the usual small-system limitation alone: periodic pyrochlore clusters contain ice-preserving ring exchanges that close only by winding around the torus. The dominant artifact is a four-site winding loop generated at order $J_{\pm}^2$, whereas the physical QSI photon ring exchange is the contractible six-site hexagon generated at order $|J_{\pm}|^3$. These notes derive the transported-dipole label

$$\delta = \rho + NL,$$

show why ordinary boundary-twist averaging does not remove the artifact, and prove two exact structure theorems that collapse the character-average protocol into its final form: at the minimal grid the character deformation is unitarily equivalent to the bare Hamiltonian through the diagonal polarization unitary $e^{2i\theta \cdot \hat{\mathbf{X}}}$, so the entire eight-point character average of the downfolded ice-band operator equals a single untwisted downfolding followed by a *polarization-parity mask*. Homotopy fixes the scope: the torus is a $K(\mathbb{Z}^3, 1)$, the winding class is the complete topological invariant of a closed process, and the mask is the maximal topologically defined cleaning. The production protocol at any $J_{\pm}$ is then *graded Brillouin–Wigner downfolding*: the exact ice-block resolvent operator $F(z) = PHP + PHQ(z - QHQ)^{-1}QHP$, masked entry by entry, solved self-consistently. It requires no eigenvectors, no band selection, and no Gram matrix; its diagnostics remain meaningful at every coupling, and its breakdown modes are physical (dissolution of the gauge sector into the spinon continuum), not numerical. Every claim is verified against exact microscopic ED on the 16-site cluster.

Contents

1	The idea in one page (for readers new to spin ice)	2
2	The physical question	3
3	Ice variables and the energy cost of defects	4
4	Degenerate perturbation theory in the ice manifold	5
5	Why the physical QSI ring exchange is a hexagon	5
5.1	Third-order hexagon amplitude	7
6	Why a four-site loop is a finite-size artifact	8
6.1	Second-order four-loop amplitude	9

7	Independent numerical verification	10
7.1	Geometry census	11
8	Why the low-temperature specific heat peak is displaced	11
9	Why ordinary boundary-twist averaging fails	12
10	The transported dipole and the zero-transport theorem	15
11	The microscopic spine: structure theorems and the polarization-parity mask	18
11.1	Gauge-theory meaning	18
11.2	Character-deformed microscopic Hamiltonian	19
11.3	Structure theorems: the character average is a mask	20
11.4	Pullback of the QED low band	21
11.5	The clean operator	22
11.6	Exact implementation checklist (Stage A)	23
12	Validation of the QED spine on cubic-16	23
12.1	Choosing the character	23
12.2	QED low-band results	24
12.3	Rejected shortcuts	24
13	Graded Brillouin–Wigner downfolding: the protocol at any $J_{\pm}$	25
13.1	Equivariance and the masked operator	26
13.2	Self-consistent solution and built-in diagnostics	26
14	The cubic-16 validation ladder	26
15	All-temperature consistency check on cubic-16	27
16	FCC-32 production protocol	30

1 The idea in one page (for readers new to spin ice)

Before any equations, here is the whole story in words and pictures.

- **What the material is.** “Quantum spin ice” (QSI) is a magnet in which the atomic magnetic moments sit on a lattice of corner-sharing tetrahedra (the *pyrochlore* lattice). On every tetrahedron the moments obey a local rule: two point in and two point out (Fig. 1). This is the same counting rule that governs the proton positions in ordinary water ice, hence the name. There are astronomically many ways to satisfy it, so the system has a huge ground-state degeneracy.
- **What makes it “quantum”.** A small quantum-mechanical term lets pairs of neighbouring moments flip. Acting over and over, these flips let the system tunnel between different two-in–two-out patterns. The smallest tunnelling move that respects the ice rule everywhere is a *six-spin ring exchange* around a hexagon (Fig. 2 and Fig. 3). Remarkably, the collective dynamics of these ring flips is described by an emergent *light*: QSI hosts its own artificial photon. This is what makes the material exciting, and it is what we want to compute.

- **Why the computation is hard.** We solve the model on a computer by exact diagonalization (ED), which requires a *finite* block of the lattice with periodic boundaries—mathematically a doughnut, or *torus*. On a torus, some sequences of flips close up not because they form a real loop in space, but only because they run off one face and come back on the opposite face (Fig. 4). The shortest such “wrap-around” move involves only *four* spins, and—crucially—it appears at a lower order in perturbation theory than the physical six-spin hexagon. It is a pure artifact of the finite box, yet it is numerically *larger* than the physics we are after (Fig. 6).
- **The symptom.** Left untreated, this artifact puts the low-temperature specific-heat peak—one of the cleanest experimental fingerprints of the emergent photon—in the wrong place, at the artifact scale $g_4 \propto J_{\pm}^2$ instead of the photon scale $g_{\text{hex}} \propto |J_{\pm}|^3$ (Figs. 7 and 8).
- **The cure.** The physical hexagon closes in real space; the artifact only closes through the wall of the box. We give a single gauge-invariant number, the *transported dipole* $\delta = \rho + \mathbf{N}\mathbf{L}$, that is exactly zero for real-space processes and nonzero for wrap-around ones (Fig. 11). In perturbation theory this says to keep only the zero-transport rows. Nonperturbatively, two exact theorems reduce the corresponding operation to something almost embarrassingly simple: express the downfolded low-energy operator in the ice basis and *delete every matrix element whose polarization difference is odd* (the parity mask). At any coupling, the object to mask is the exact Brillouin–Wigner ice-block operator $F(z)$, which exists whether or not the low band is still ice-like. We also show why simpler fixes, such as boundary-twist averaging or averaging thermodynamic curves, do not remove the artifact (Fig. 10) — indeed for thermodynamic curves we prove that they *cannot*.

The rest of these notes makes each of these statements precise and checks every number with an independent, from-scratch script.

2 The physical question

The QSI Hamiltonian considered here is the nearest-neighbor XXZ model in local pyrochlore frames,

$$H = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- + S_i^- S_j^+), \quad |J_{\pm}| \ll J_{zz}, \quad J_{zz} > 0. \quad (1)$$

The Ising part has an extensively degenerate ground manifold: every tetrahedron has two spins pointing in and two spins pointing out. This is the ice manifold $\mathcal{I}$.

For $J_{\pm} > 0$, the off-diagonal matrix elements in the S^z basis are non-positive, and standard QMC can be sign-problem-free. For $J_{\pm} < 0$, the off-diagonal matrix elements have the opposite sign. This is the frustrated π -flux QSI sector. It is precisely the sector where ED is most valuable, because QMC no longer gives an unbiased finite-temperature benchmark.

The central difficulty is that ED necessarily uses a finite periodic cluster. Such a cluster is a torus. On a torus, paths can close by periodic identification even when they do not close in real space. Some of these winding paths preserve the ice rule and therefore appear directly in the low-energy gauge Hamiltonian. The shortest such processes on the standard 16-site cubic and 32-site FCC clusters are four-site loops. They are not physical bulk plaquettes. Nevertheless, they strongly affect the low-temperature specific heat and the low-frequency $S^{zz}(\mathbf{q}, \omega)$.

The ice rule: every tetrahedron wants two spins in and two out

two-in / two-out

three-in / one-out

one-in / three-out

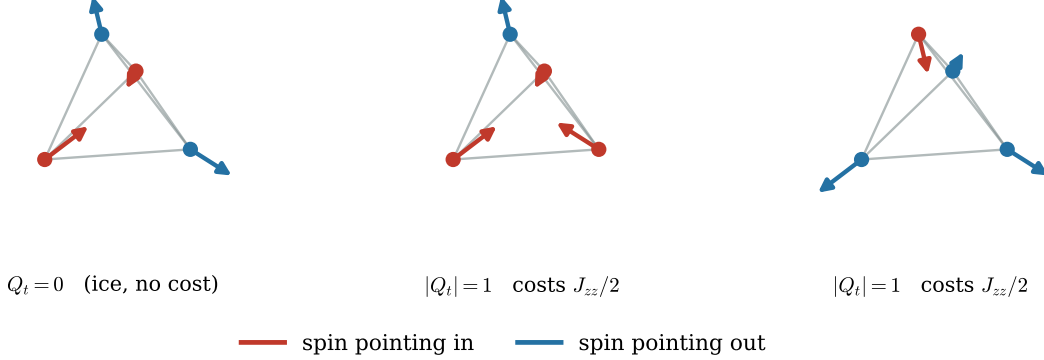


Figure 1: **The ice rule.** Each spin lives on a corner of a tetrahedron and points either into the tetrahedron (red) or out of it (blue). *Left:* the low-energy “ice” configuration has two in and two out, so the net Ising charge $Q_t = \sum_{i \in t} S_i^z$ vanishes and the tetrahedron pays no energy. *Middle, right:* tipping one spin makes the tetrahedron three-in-one-out or one-in-three-out, with $|Q_t| = 1$; by Eq. (3) this costs $J_{zz}/2$. These charged tetrahedra are the “spinons” of quantum spin ice.

3 Ice variables and the energy cost of defects

For a tetrahedron t , define the Ising charge

$$Q_t = \sum_{i \in t} S_i^z. \quad (2)$$

For spin $1/2$, $S_i^z = \pm 1/2$. The Ising energy on one tetrahedron is

$$\sum_{\langle ij \rangle \in t} S_i^z S_j^z = \frac{1}{2} \left[\left(\sum_{i \in t} S_i^z \right)^2 - \sum_{i \in t} (S_i^z)^2 \right] = \frac{1}{2} Q_t^2 - \frac{1}{2}. \quad (3)$$

Thus, up to a constant,

$$H_0 = \frac{J_{zz}}{2} \sum_t Q_t^2. \quad (4)$$

The ice rule is $Q_t = 0$ on every tetrahedron. A 3-in-1-out or 1-in-3-out tetrahedron has $|Q_t| = 1$ and costs $J_{zz}/2$ relative to an ice tetrahedron.

Now act with a transverse exchange

$$S_i^+ S_j^- + S_i^- S_j^+$$

on an antiparallel nearest-neighbor pair. The two sites i and j lie on one common tetrahedron, where exchanging an up and a down spin preserves $Q_t = 0$. Each site also belongs to one other tetrahedron. Those two neighboring tetrahedra become charged, each costing $J_{zz}/2$. Therefore a single nearest-neighbor exchange out of the ice manifold creates two defects with total cost

$$\Delta_{1\text{hop}} = J_{zz}. \quad (5)$$

This simple denominator is the origin of the perturbative scales below.

In plain language

A single spin flip is like pushing one domino: it necessarily unbalances the two tetrahedra that share that spin, creating a pair of oppositely charged defects that together cost J_{zz} . To get back into the ice manifold you must flip more spins so that every tetrahedron is balanced again. The cheapest way to do that around a closed path is what the next sections count.

4 Degenerate perturbation theory in the ice manifold

Let P be the projector onto the ice manifold and $Q = 1 - P$. Write

$$H = H_0 + V, \quad V = -J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- + S_i^- S_j^+).$$

The ice energy is shifted to zero. The non-ice resolvent is

$$R = Q \frac{1}{E_{\mathcal{I}} - H_0} Q = -Q H_0^{-1} Q. \quad (6)$$

Because a single exchange leaves the ice manifold, $PVP = 0$. The first nonzero terms are therefore

$$H_{\text{eff}}^{(2)} = PVRVP, \quad (7)$$

$$H_{\text{eff}}^{(3)} = PVRV RVP. \quad (8)$$

For a path $s \rightarrow m_1 \rightarrow \dots \rightarrow t$, its contribution has the form

$$\langle t | H_{\text{eff}}^{(k)} | s \rangle = \sum_{\text{paths}} \frac{\langle t | V | m_{k-1} \rangle \dots \langle m_1 | V | s \rangle}{(E_{\mathcal{I}} - E_{m_1}) \dots (E_{\mathcal{I}} - E_{m_{k-1}})}. \quad (9)$$

Each matrix element of V contributes a factor $-J_{\pm}$, and each intermediate energy denominator is negative.

Remark 1. The standalone recomputation stores each perturbative process as a row

$$r = (s, t, \mathbf{N}, c, k),$$

meaning that the row connects ice states s and t , appears at order k , has coefficient c when $J_{\pm} = 1$, and has accumulated periodic image transport $\mathbf{N}$. At physical coupling, the row contributes $cJ_{\pm}^k |t\rangle\langle s|$.

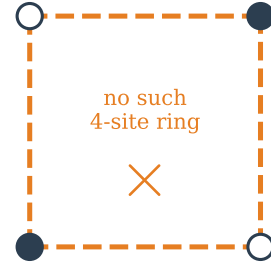
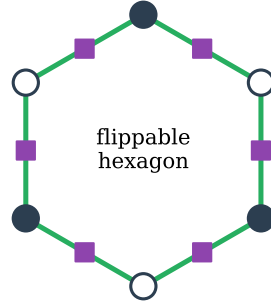
5 Why the physical QSI ring exchange is a hexagon

The pyrochlore lattice is the line graph of the diamond lattice: a pyrochlore site corresponds to a diamond bond, and a pyrochlore tetrahedron corresponds to a diamond vertex. An ice configuration assigns a divergence-free electric field to diamond bonds. A ring exchange in the ice manifold must flip an alternating set of spins around a closed path so that the divergence at every touched diamond vertex remains zero.

Why the smallest physical ring exchange has six spins

Allowed: 6-site ring

Forbidden in the bulk: 4-site ring



diamond lattice is bipartite $\Rightarrow$ shortest loop has 6 links
 tetrahedron (diamond vertex)



a 4-cycle needs two same-colour sites adjacent
 (a diamond 4-loop) — it does not exist
 spin (pyrochlore site = diamond link)

Figure 2: **The smallest real ring exchange has six spins.** Each tetrahedron is a vertex of the diamond lattice (circles); each spin is a link of that lattice (squares). A ring exchange must flip an alternating chain of spins around a closed loop of tetrahedra. *Left:* the diamond lattice is bipartite (two colours, links only join different colours), so its shortest loop uses six links—a six-spin hexagon. *Right:* a four-spin ring would require a four-loop on the diamond lattice, which would need two same-colour vertices to be adjacent; no such loop exists in the bulk. This is the geometric reason a contractible four-site ice loop is impossible.

Lemma 1 (Shortest contractible ice loop). *The shortest nontrivial contractible ice-preserving ring exchange on the infinite pyrochlore lattice is a six-site hexagon.*

Proof. An ice-preserving loop must enter and leave every touched tetrahedron, so on the diamond lattice it corresponds to a closed non-backtracking loop. The diamond lattice is bipartite and has no triangles. It also has no four-cycles; its elementary loops are length six. Therefore the shortest contractible closed diamond loop has length six. Mapping back to the pyrochlore line graph gives a six-site pyrochlore hexagon. A triangle of the pyrochlore graph lies inside one tetrahedron and is not an alternating ice ring. A contractible four-site ice ring would map to a diamond four-cycle, which does not exist. $\square$

5.1 Third-order hexagon amplitude

The physical ring exchange: amplitude $g_{\text{hex}} = 12|J_{\pm}|^3/J_{zz}^2$ (third order in $J_{\pm}$)

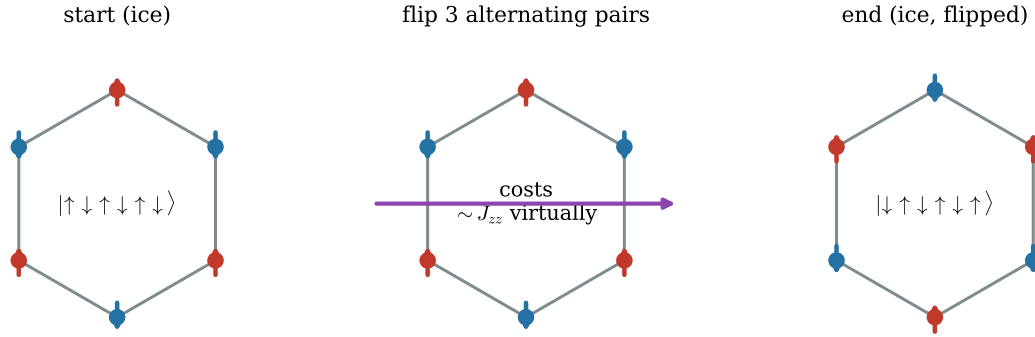


Figure 3: **The physical ring exchange (the emergent-photon move).** Starting from an alternating up/down pattern on a hexagon, flipping all three alternating pairs turns each up into a down and vice versa, returning to the ice manifold. Because the intermediate steps leave the ice manifold, the amplitude is third order in $J_{\pm}$, with magnitude $g_{\text{hex}} = 12|J_{\pm}|^3/J_{zz}^2$. The coherent superposition of these ring flips across the lattice is the artificial photon of quantum spin ice.

Consider a flippable hexagon with alternating spins. To move from one ice configuration to the flipped configuration, one must exchange three alternating nearest-neighbor pairs. After the first exchange the system has two charged tetrahedra and costs J_{zz} . After the second exchange it still has a pair of charged tetrahedra and costs J_{zz} . Thus each ordering gives a denominator J_{zz}^2 .

For one orientation there are $3!$ orderings. Including the Hermitian conjugate orientation gives the conventional ring term

$$H_{\text{eff}}^{\text{hex}} = -g_{\text{hex}} \sum_{\text{hex}} (S_1^+ S_2^- S_3^+ S_4^- S_5^+ S_6^- + \text{h.c.}), \quad (10)$$

with magnitude

$$g_{\text{hex}} = \frac{12|J_{\pm}|^3}{J_{zz}^2}. \quad (11)$$

The sign of the effective ring term follows $\text{sgn}(J_{\pm}^3)$. This sign is what distinguishes the 0-flux and π -flux gauge sectors.

6 Why a four-site loop is a finite-size artifact

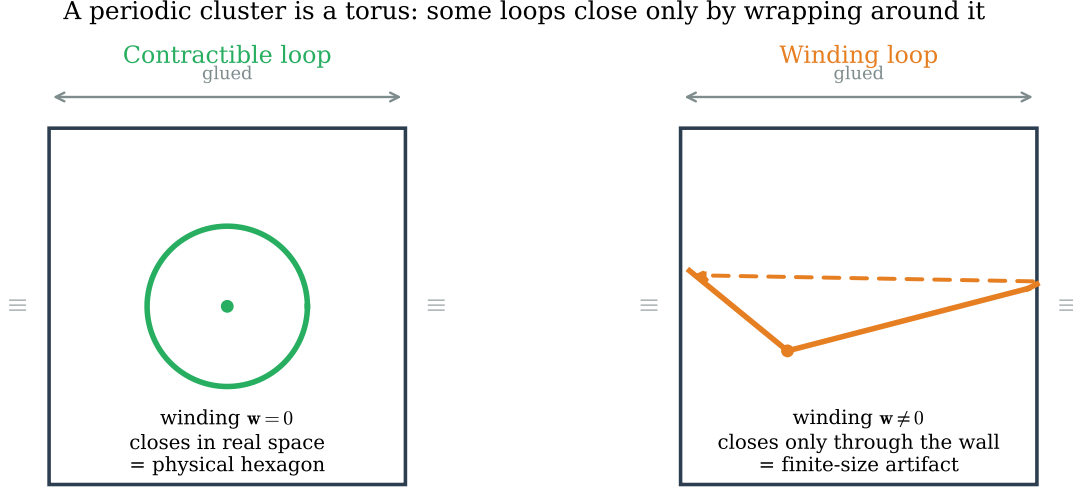


Figure 4: **A periodic cluster is a torus.** Opposite walls of the simulation box are identified (glued), so the box is topologically a doughnut. *Left:* a loop that closes inside the box has winding $w = 0$; it is a genuine real-space process such as the physical hexagon. *Right:* a path can also close by leaving one wall and re-entering the opposite wall. Its winding $w \neq 0$; it exists *only* because of the periodic gluing and has no counterpart in the infinite lattice.

A periodic cluster is not $\mathbb{R}^3$; it is the flat torus

$$T^3 = \mathbb{R}^3 / (\mathbb{Z}^3 \mathbf{L}), \quad (12)$$

where we use row vectors and the rows of $\mathbf{L}$ are the three cluster translation vectors. Thus $\mathbf{mL}$, with $\mathbf{m} \in \mathbb{Z}^3$, is a periodic translation. For the cubic $L_1 \times L_2 \times L_3$ cluster,

$$\mathbf{L} = \begin{pmatrix} L_1 & 0 & 0 \\ 0 & L_2 & 0 \\ 0 & 0 & L_3 \end{pmatrix}.$$

For the FCC primitive cluster, the rows are $L_1(\frac{1}{2}, \frac{1}{2}, 0)$, $L_2(\frac{1}{2}, 0, \frac{1}{2})$, and $L_3(0, \frac{1}{2}, \frac{1}{2})$. All coordinates in this note are measured in conventional cubic-cell units. The pyrochlore basis sites lie on the quarter-lattice,

$$\mathbf{r}_i \in \frac{1}{4}\mathbb{Z}^3,$$

which is the reason factors of 4 appear below.

A bond from site i to site j is represented by a minimum-image displacement

$$\mathbf{d}_{ij} = \mathbf{r}_j - \mathbf{r}_i - \mathbf{n}_{ij}\mathbf{L}, \quad \mathbf{n}_{ij} \in \mathbb{Z}^3. \quad (13)$$

For a closed path $\mathcal{C}$ on the periodic graph, define its winding vector

$$\mathbf{w}_{\mathcal{C}} = \sum_{(ij) \in \mathcal{C}} \mathbf{n}_{ij}. \quad (14)$$

If $w_C = 0$, the lifted path closes in real space and is contractible. If $w_C \neq 0$, the path closes only because opposite faces of the cluster have been identified.

Proposition 1 (No physical four-loop). *Every ice-preserving four-cycle on the standard $1 \times 1 \times 1$ cubic and $2 \times 2 \times 2$ FCC pyrochlore clusters is a winding torus loop, not a contractible bulk process.*

Proof. The geometric lemma above rules out a contractible four-site ice loop on the infinite pyrochlore lattice. Therefore any four-site ice loop observed on a periodic cluster must use the periodic identification. Equivalently, its lift to $\mathbb{R}^3$ fails to close and has nonzero winding w_C . The explicit independent enumeration below verifies this for the two clusters used in the ED calculation. $\square$

6.1 Second-order four-loop amplitude

Each transverse exchange costs one power of $J_\pm$ and climbs J_{zz} ;
one extra rung makes the hexagon a weaker, higher-order process

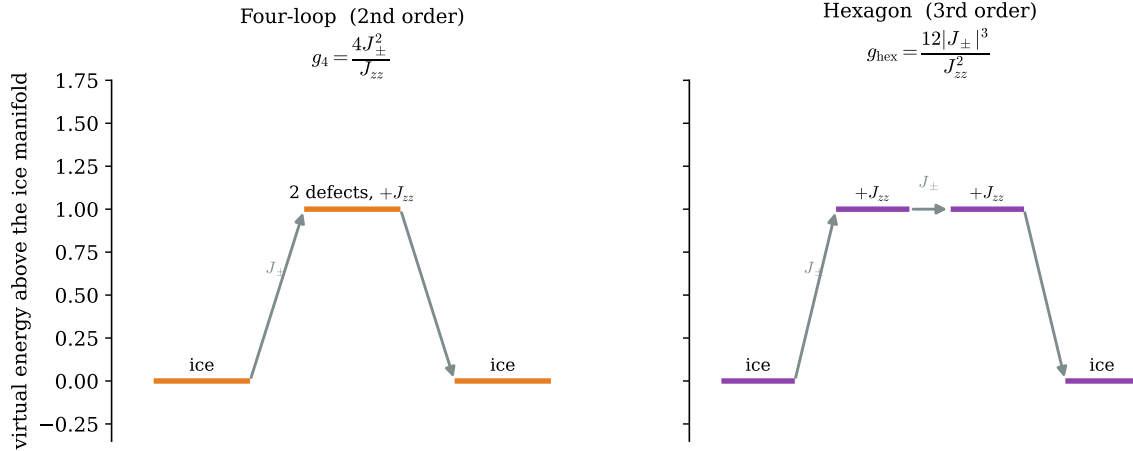


Figure 5: **Counting the powers of $J_\pm$.** Every transverse exchange adds one factor of $J_\pm$ and pushes the system one virtual step above the ice manifold, where it costs $\sim J_{zz}$. *Left:* the winding four-loop returns to the ice manifold after just two exchanges (one intermediate tetrahedron pair), giving $g_4 = 4J_\pm^2/J_{zz}$. *Right:* the hexagon needs three exchanges (two intermediate steps), giving the weaker, higher-order $g_{\text{hex}} = 12|J_\pm|^3/J_{zz}^2$. The extra rung is the entire reason the artifact is parametrically larger at small $J_\pm$.

A winding four-loop has four alternating sites and can be flipped by two nearest-neighbor exchanges. The two exchanged bonds form a perfect matching of the four sites. There are two matchings, and for each matching there are two time orderings. Each path has one intermediate state with energy cost J_{zz} . Thus the magnitude of the second-order matrix element is

$$4 \frac{J_\pm^2}{J_{zz}}. \quad (15)$$

In the ring-exchange convention used here,

$$\boxed{g_4 = \frac{4J_\pm^2}{J_{zz}}}. \quad (16)$$

The sign is convention dependent, but the scale is not.

The scale separation is

$$\frac{g_4}{g_{\text{hex}}} = \frac{4J_{\pm}^2/J_{zz}}{12|J_{\pm}|^3/J_{zz}^2} = \frac{J_{zz}}{3|J_{\pm}|}. \quad (17)$$

Thus the artifact becomes more severe precisely in the perturbative regime where the QSI description is best controlled.

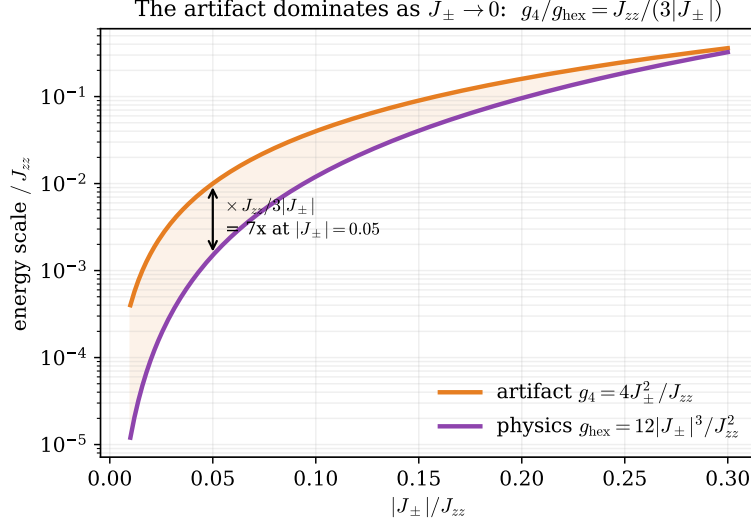


Figure 6: **The artifact dominates exactly where we want to work.** The finite-size four-loop scale g_4 (orange) and the physical photon scale g_{hex} (purple) both vanish as $J_{\pm} \rightarrow 0$, but their ratio $g_4/g_{\text{hex}} = J_{zz}/(3|J_{\pm}|)$ grows. At the representative value $|J_{\pm}| = 0.05 J_{zz}$ the artifact is already $\sim 7\times$ larger than the physics it would otherwise swamp.

7 Independent numerical verification

The script

`notes/recompute_finite_size_artifact.py`

does not import the old `twist_qsi_demo/scripts` code and does not inspect git history. It independently:

- (i) builds the pyrochlore positions, nearest-neighbor graph, tetrahedra, and periodic image vectors;
- (ii) enumerates all two-in-two-out ice states;
- (iii) enumerates ice-preserving four-cycles and hexagons;
- (iv) constructs $H_2 = PVRVP$ and $H_3 = PVRVRVP$ from path sums;
- (v) applies either no projector or the zero-transport projector derived below;
- (vi) diagonalizes the resulting ice-manifold Hamiltonian and computes $C(T)$.

The raw output is saved in `notes/recomputed_finite_size_artifact.json`.

7.1 Geometry census

cluster	sites	ice states	4-cycles	contractible 4-cycles	contractible hexagons
cubic $1 \times 1 \times 1$	16	90	36	0	16
FCC $2 \times 2 \times 2$	32	2970	24	0	32

The same enumeration finds 48 wrapping hexagons on the 16-site cluster and 96 wrapping hexagons on the 32-site cluster. Those are also finite-size torus processes, but they appear at the same order as the physical hexagons. The four-loops are the dominant contamination because they appear one order earlier.

8 Why the low-temperature specific heat peak is displaced

For an exact spectrum $\{E_n\}$, shifted so $E_0 = 0$,

$$Z(T) = \sum_n e^{-E_n/T}, \quad \langle E^p \rangle_T = \frac{1}{Z} \sum_n E_n^p e^{-E_n/T}, \quad (18)$$

and

$$C(T) = \frac{\langle E^2 \rangle_T - \langle E \rangle_T^2}{T^2}. \quad (19)$$

For a two-level system with splitting Δ ,

$$C_{2\text{lev}}(T) = \frac{(\Delta/T)^2 e^{-\Delta/T}}{(1 + e^{-\Delta/T})^2}, \quad (20)$$

whose maximum is at $T = O(\Delta)$. A many-level finite cluster has a more complicated spectrum, but the lower peak still tracks the dominant low-energy splitting scale in the ice manifold.

Therefore:

$$T_{\text{peak}}^{\text{bare}} \sim g_4 \propto J_{\pm}^2, \quad \text{if winding four-loops are present,} \quad (21)$$

$$T_{\text{peak}}^{\text{clean}} \sim g_{\text{hex}} \propto |J_{\pm}|^3, \quad \text{after winding transport is removed.} \quad (22)$$

The independent calculation gives the following representative peaks:

cluster	Hamiltonian	$J_{\pm}$	T_{peak}	scale diagnostic
cubic $N = 16$	bare	-0.05	0.010056	$T_{\text{peak}}/g_4 = 1.01$
cubic $N = 16$	clean	-0.05	0.001604	$T_{\text{peak}}/g_{\text{hex}} = 1.07$
FCC $N = 32$	bare	-0.05	0.004180	$T_{\text{peak}}/g_{\text{hex}} = 2.79$
FCC $N = 32$	clean	-0.05	0.000782	$T_{\text{peak}}/g_{\text{hex}} = 0.52$
FCC $N = 32$	bare	-0.10	0.018053	$T_{\text{peak}}/g_{\text{hex}} = 1.50$
FCC $N = 32$	clean	-0.10	0.006252	$T_{\text{peak}}/g_{\text{hex}} = 0.52$

For the 16-site cluster at $J_{\pm} = -0.05$, $g_4 = 4(0.05)^2 = 0.010$ and $g_{\text{hex}} = 12(0.05)^3 = 0.0015$, so the scale identification is especially transparent.

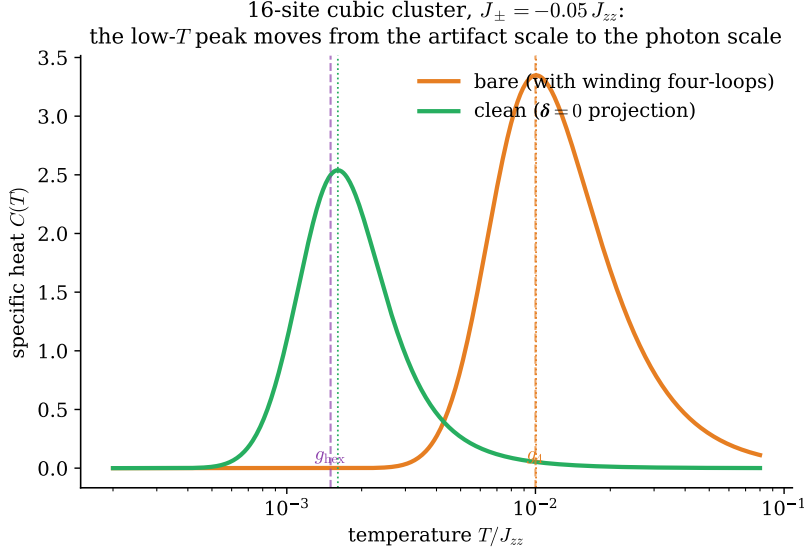


Figure 7: **The low-temperature specific-heat peak is displaced by the artifact.** Exact specific heat of the 16-site cubic ice-manifold Hamiltonian at $J_{\pm} = -0.05 J_{zz}$. The bare cluster (orange) peaks at the artifact scale g_4 ; after the $\delta = 0$ projection (green) the peak drops onto the physical photon scale g_{hex} —nearly an order of magnitude lower in temperature. Since the specific heat is a nonlinear function of the spectrum, this shift cannot be undone by averaging $C(T)$ after the fact (Sec. 11).

9 Why ordinary boundary-twist averaging fails

Boundary twists are often used to reduce finite-size effects. Put phases only on boundary-crossing bonds. If a virtual path accumulates image transport $\mathbf{N}$, its row coefficient becomes

$$c_r(\phi) = c_r e^{-i\mathbf{N}_r \cdot \phi}. \quad (23)$$

The eight-corner average over $\phi_a \in \{0, \pi\}$ gives

$$\frac{1}{8} \sum_{\phi_a=0,\pi} e^{-i\mathbf{N} \cdot \phi} = \prod_{a=1}^3 \frac{1 + (-1)^{N_a}}{2}. \quad (24)$$

This keeps rows with even $\mathbf{N}$.

The subtle point is that the geometric ring exchange is not a single path. It is a sum over virtual paths. A winding four-loop has multiple matchings and time orderings. Some paths cross the boundary, while other paths implementing the same geometric flip can have $\mathbf{N} = 0$ in the boundary gauge. The corner average therefore kills only part of the coefficient.

The standalone recomputation classifies the off-diagonal rows by geometric channel and gives:

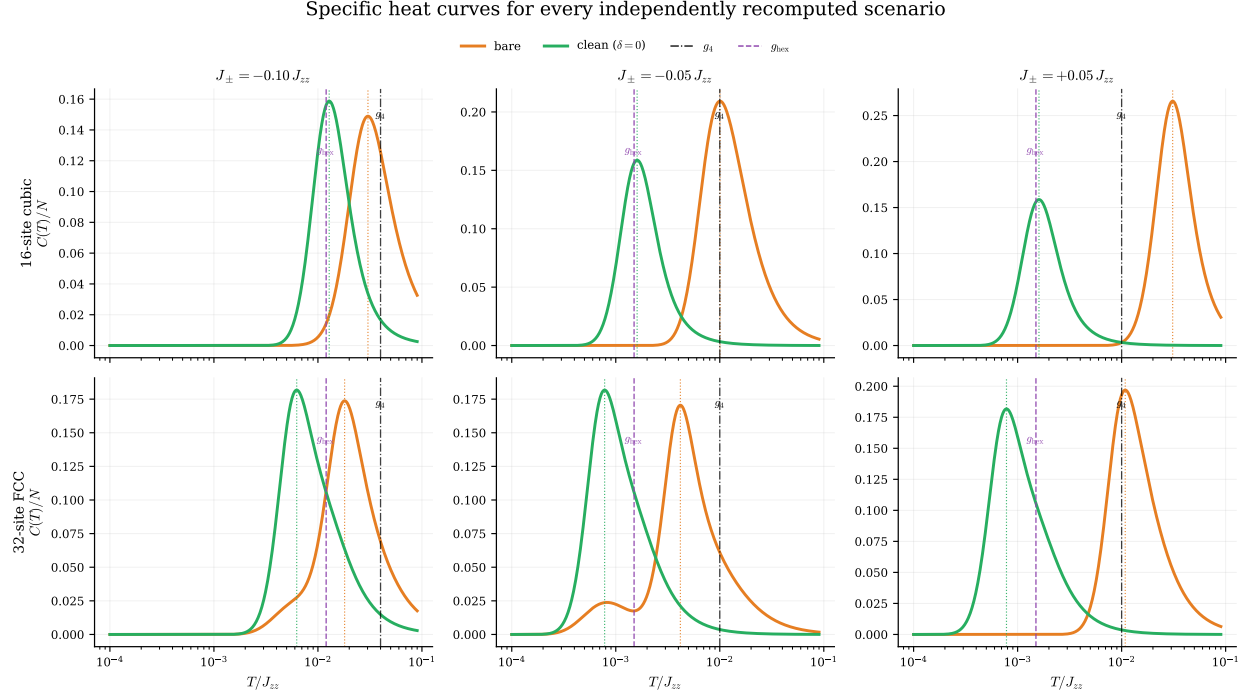


Figure 8: **Specific heat as a function of temperature for all independently recomputed scenarios.** Each panel shows the exact ice-manifold specific heat per site, $C(T)/N$, from the standalone Schrieffer–Wolff calculation. The columns are the three computed couplings, and the rows are the two ED clusters. Orange is the bare periodic effective Hamiltonian; green is the zero-transport ($\delta = 0$) Hamiltonian. Dotted orange/green vertical lines mark the actual peak positions of the two curves, the black dash-dot line marks g_4 , and the purple dashed line marks g_{hex} . The full curves show the same conclusion as the peak table: removing transported winding processes shifts the low-temperature feature from the finite-size scale toward the photon scale.

All independently recomputed finite-size scenarios

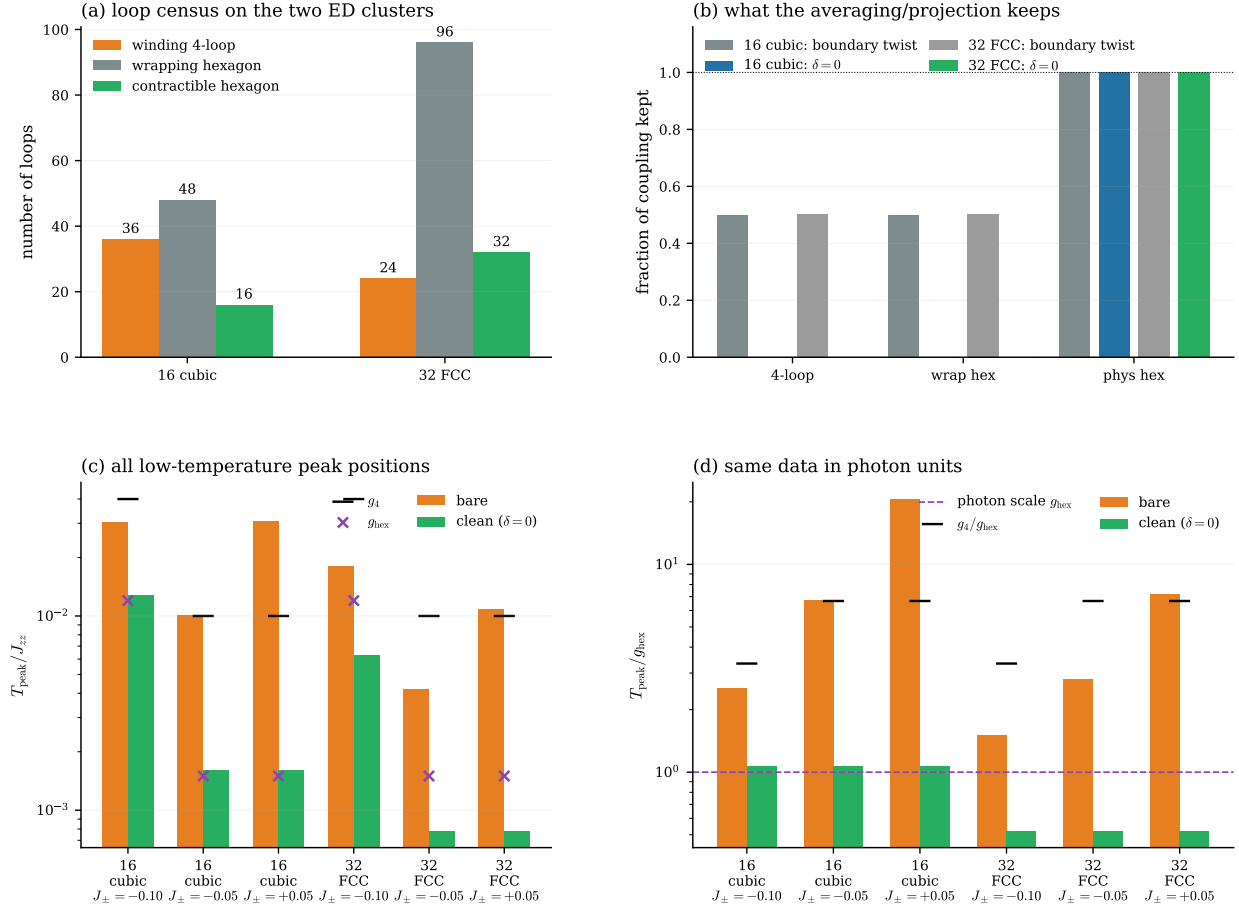


Figure 9: **All independently recomputed finite-size scenarios in one place.** (a) The two ED clusters both contain winding four-loops and wrapping hexagons in addition to the physical contractible hexagons. (b) Boundary-twist averaging keeps half of each winding channel on both clusters, while the zero-transport projector removes them and keeps the physical hexagon. (c) The low-temperature peak positions for all computed clusters, flux signs, and projection choices. Black ticks mark g_4 and purple crosses mark g_{hex} . (d) The same peak data in photon units. The cleaned spectra stay at an order-one multiple of g_{hex} , while the bare spectra are pushed upward by the winding-loop contamination.

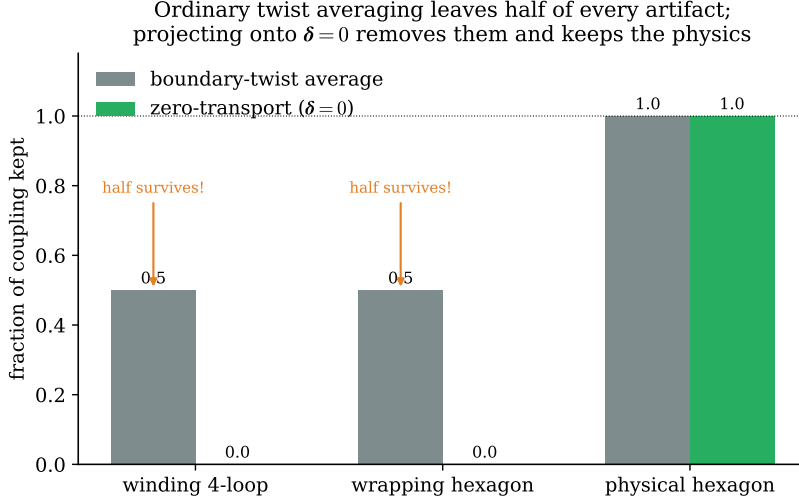


Figure 10: **Boundary-twist averaging is not enough.** Fraction of each geometric channel’s coupling that survives on the 16-site cubic cluster. Ordinary boundary-twist (corner) averaging (grey) leaves *half* of every winding artifact—because the ring exchange is a sum over virtual paths and some of those paths do not cross the boundary in the chosen gauge. The gauge-invariant $\delta = 0$ projection (green) removes the winding four-loop and wrapping hexagon completely while keeping the physical hexagon at full strength.

channel	corner average kept	$N = 0$ kept	zero-transport kept
4-loop, cubic $N = 16$	0.5	0.5	0
wrapping hexagon, cubic $N = 16$	0.5	0.5	0
contractible hexagon, cubic $N = 16$	1	1	1
4-loop, FCC $N = 32$	0.5	0.5	0
wrapping hexagon, FCC $N = 32$	0.5	0.5	0
contractible hexagon, FCC $N = 32$	1	1	1

Thus ordinary boundary-twist averaging reduces the winding artifact but does not remove it.

10 The transported dipole and the zero-transport theorem

The correct object is not the boundary-crossing count of one chosen gauge. It is the gauge-invariant net displacement transported by the effective operator.

Definition 1 (Home-cell dipole). For a perturbative row connecting ice states s and t , define

$$R_+(t, s) = \{i : S_i^z(t) - S_i^z(s) > 0\}, \quad R_-(t, s) = \{j : S_j^z(t) - S_j^z(s) < 0\}.$$

The set R_+ contains spins raised by the effective process; R_- contains spins lowered by the effective process. The home-cell dipole is

$$\rho_{t,s} = \sum_{i \in R_+(t,s)} \mathbf{r}_i - \sum_{j \in R_-(t,s)} \mathbf{r}_j. \quad (25)$$

Definition 2 (Transported dipole). Let $\mathbf{N}_r \in \mathbb{Z}^3$ be the accumulated image vector of the virtual path. In row vector convention, $\mathbf{N}_r \mathbf{L}$ is the net periodic translation needed when the path is lifted from the torus back to the covering space $\mathbb{R}^3$. The transported dipole is

$$\boxed{\boldsymbol{\delta}_r = \boldsymbol{\rho}_{t,s} + \mathbf{N}_r \mathbf{L}} \quad (26)$$

The formula is easiest to understand for one microscopic hop. The operator $S_i^+ S_j^-$ raises site i and lowers site j , so its home-cell dipole is $\mathbf{r}_i - \mathbf{r}_j$. If the nearest-neighbor bond was represented using the periodic image vector $\mathbf{n}_{ij}$, then the lifted displacement of the spin transfer is

$$\mathbf{r}_i - \mathbf{r}_j + \mathbf{n}_{ij} \mathbf{L} = -\mathbf{d}_{ij}.$$

The many-step perturbative row adds these lifted increments. Thus $\boldsymbol{\delta}_r$ is not a new dynamical charge. It is the net real-space displacement carried by the virtual spin transfers after the periodic images have been undone.

For the order- $J_{\pm}^2$ and order- $J_{\pm}^3$ ice-manifold rows explicitly enumerated below, the possible transported dipoles are coarser than the microscopic quarter-lattice: their components lie in $\frac{1}{2}\mathbb{Z}$. This is the only set of virtual processes that controls the unwanted four-loop peak and the physical hexagon peak in the present campaign.

Both loops flip alternating spins and preserve the ice rule — only their torus transport differs

Contractible hexagon (physics)

Winding four-loop (artifact)

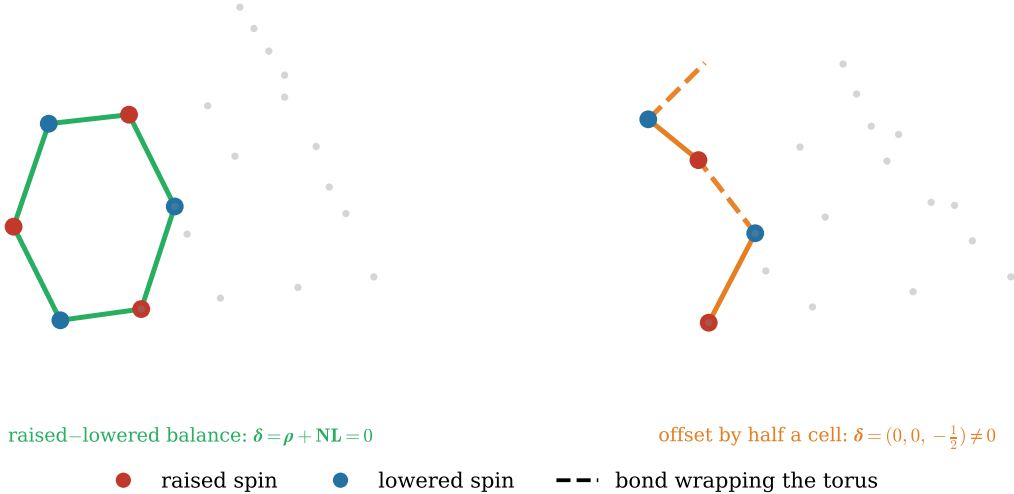


Figure 11: **The transported dipole tells the two loops apart, on the real 16-site cubic cluster.** Both loops flip alternating spins (red raised, blue lowered) and both return the system to the ice manifold. *Left:* the contractible hexagon is balanced—the raised and lowered spins average to the same point—so its transported dipole vanishes, $\boldsymbol{\delta} = \boldsymbol{\rho} + \mathbf{N}\mathbf{L} = 0$. *Right:* the winding four-loop must cross the torus wall (dashed bonds); its raised and lowered spins are offset by half a unit cell, giving $\boldsymbol{\delta} = (0, 0, -\frac{1}{2}) \neq 0$. Selecting $\boldsymbol{\delta} = 0$ keeps the left process and discards the right one.

Lemma 2 (Gauge invariance of δ). *The vector δ_r is independent of the choice of home cell and of the gauge used to place the twist phases.*

Proof. Move a site coordinate by a cluster translation, $\mathbf{r}_i \mapsto \mathbf{r}_i + \mathbf{m}\mathbf{L}$ with $\mathbf{m} \in \mathbb{Z}^3$. If that site is raised, ρ changes by $+\mathbf{m}\mathbf{L}$; if it is lowered, ρ changes by $-\mathbf{m}\mathbf{L}$. The image bookkeeping of the virtual path changes by the opposite integer vector, because the same physical hop is now represented using a different periodic image. Therefore $\mathbf{N}\mathbf{L}$ changes by the negative of the change in ρ , and the sum $\rho + \mathbf{N}\mathbf{L}$ is unchanged. $\square$

Proposition 2 (Transported-dipole character projection). *For the order- $J_\pm^2$ and order- $J_\pm^3$ ice-manifold rows on the clusters considered here, every transported dipole has components in $\frac{1}{2}\mathbb{Z}$. We therefore define the integer row label*

$$\boxed{\Delta_r \equiv 2\delta_r \in \mathbb{Z}^3.} \quad (27)$$

This is only a change of units for these rows. For example, the winding four-loop in Fig. 11 has $\delta = (0, 0, -\frac{1}{2})$, hence $\Delta = (0, 0, -1)$. The condition $\Delta = 0$ is exactly the same condition as $\delta = 0$.

Couple an auxiliary character field θ to this integer transported dipole. A perturbative row r acquires the phase

$$c_r(\theta) = c_r e^{-i\theta \cdot \Delta_r}. \quad (28)$$

The operator-level average over $\theta \in [0, 2\pi)^3$ keeps exactly the rows with $\delta_r = 0$.

Proof. The characters of the abelian group $\mathbb{Z}^3$ are orthogonal:

$$\overline{c_r} = c_r \int_{[0, 2\pi)^3} \frac{d^3\theta}{(2\pi)^3} e^{-i\theta \cdot \Delta_r} = c_r \delta_{\Delta_r, 0}.$$

Since $\Delta_r = 2\delta_r$, the condition $\Delta_r = 0$ is equivalent to $\delta_r = 0$. $\square$

Remark 2 (How many discrete character points are needed?). The exact integral over θ can be replaced by a finite Fourier sum if the grid separates the transported-dipole sectors present in the calculation. For one component,

$$\frac{1}{M} \sum_{m=0}^{M-1} e^{-i(2\pi m/M)\Delta} = \begin{cases} 1, & \Delta = 0 \pmod{M}, \\ 0, & \Delta \neq 0 \pmod{M}. \end{cases}$$

Thus the finite grid projects onto $\Delta = 0$ only modulo M . In the order- $J_\pm^2$ and order- $J_\pm^3$ row data, the nonzero components of $\Delta = 2\delta$ are ± 1 . Therefore the two-point grid $M = 2$ already distinguishes the finite-size sectors from the clean zero-transport sector.

This is a pure aliasing issue. Consider one spurious four-loop component with $\delta_\mu = -\frac{1}{2}$. With the effective-row label $\Delta_\mu = 2\delta_\mu = -1$, the two-point average gives

$$\frac{1}{2} \left(1 + e^{-i\pi(-1)} \right) = 0,$$

so the row is removed. The perturbative row-table check on the 16-site cluster gives exactly this:

integer label	M	relative error to strict $\delta = 0$
2δ	2	0

Thus the minimal character projection for this campaign is 2δ with $M = 2$.

Remark 3 (Why physical twists were insufficient). A physical flat vector potential gives phases of the form $e^{-i\mathbf{A}\cdot\delta_r}$. On the 16-site cluster the spurious four-loops have half-cell transported dipoles, so the corresponding exponent is half-integer in 2π units. Physical twist averaging therefore suppresses these rows but does not project them out exactly. The auxiliary character field in Eq. (28) is the finite-cluster source conjugate to the integer transported dipole $\Delta = 2\delta$.

Corollary 1 (Clean ice-manifold Hamiltonian). *The artifact-free gauge Hamiltonian is*

$$H_{\text{clean}}(J_{\pm}) = \sum_{r: \delta_r=0} c_r J_{\pm}^{k_r} |t_r\rangle \langle s_r|. \quad (29)$$

It keeps contractible hexagons and removes winding four-loops and wrapping hexagons.

11 The microscopic spine: structure theorems and the polarization-parity mask

The perturbative derivation above identifies the topological quantum number that must be removed: nonzero transported dipole. The production ED protocol does not stop at the truncated row table. Its central object is a *downfolded low-energy operator expressed in the ice basis*, and the transported-dipole character projection is applied to that operator before any thermodynamics are computed. Historically this was implemented as an eight-point character average of QED low-band extractions; the structure theorems of Sec. 11.3 collapse it to

$$H_{\text{mic}} \xrightarrow{\text{downfold at } \theta=0} H_{\text{band}} \xrightarrow{\text{parity mask}} H_{\text{band}}^{\text{clean}} \xrightarrow{\text{ED}} C(T), S(T), \text{ gauge-sector observables,}$$

a single untwisted computation followed by an exact entrywise projection. Perturbation theory supplies the character label and the weak-coupling benchmark; the microscopic solver supplies the dressed low-band operator; the mask supplies the projection.

11.1 Gauge-theory meaning

The pyrochlore lattice is the line graph of the diamond lattice. A spin on a pyrochlore site is a directed electric field on a diamond link,

$$E_{\ell} = S_i^z,$$

up to an orientation convention. The ice rule is Gauss' law,

$$(\nabla \cdot E)_r = \sum_{\ell \supset r} \eta_{r\ell} E_{\ell} = 0. \quad (30)$$

so the ice manifold is the zero-charge sector of a compact $U(1)$ gauge theory. Integrating out virtual spinons generates local magnetic dynamics,

$$H_{\text{eff}}^{\text{bulk}} = -g_{\text{hex}} \sum_{p \in \text{hexagons}} \cos[(\nabla \times A)_p] + \dots, \quad (31)$$

where A_{ℓ} is conjugate to E_{ℓ} . This is a local plaquette term: the physical magnetic ring is a contractible diamond hexagon.

On a three-torus there are also noncontractible Wilson loops, classified by

$$H_1(T^3, \mathbb{Z}) = \mathbb{Z}^3.$$

In the thermodynamic gauge theory these are global tunnelling events, not local photon dynamics. On a tiny periodic cluster, however, a noncontractible Wilson loop can be only four spins long. The protocol removes these finite-size topological tunnelling events while retaining the local contractible gauge dynamics.

Remark 4 (Homotopy fixes the scope of any cleaning scheme). The torus is aspherical: $T^3 = K(\mathbb{Z}^3, 1)$, so $\pi_n(T^3) = 0$ for $n \geq 2$ and free homotopy classes of closed loops are in bijection with $\pi_1(T^3) = H_1(T^3, \mathbb{Z}) = \mathbb{Z}^3$. Two consequences frame everything below. (i) *Completeness*. The winding class is the *only* topological invariant of a completed low-energy process; there is no torsion and no higher obstruction. A cleaning scheme keyed to topological labels can remove at most the nontrivial classes, and the transported-dipole projection removes exactly these: it is the *maximal* topologically defined cleaning. (ii) *Matched pairs are honestly trivial*. A process that winds by $+\mathbf{w}$ and later by $-\mathbf{w}$ lifts to a closed loop in the covering space $\mathbb{R}^3$; it has a genuine infinite-lattice counterpart (a virtual spinon excursion of length $\sim L$) and *cannot* be distinguished from bulk physics by any superselection label. Such contributions are ordinary finite-size image corrections, to be controlled by comparing cluster sizes, not pathologies to be projected. This resolves, as a theorem rather than an observation, why matched-winding contributions survive every averaging scheme.

At the group level, the transported dipoles generate an abelian group $\Gamma \subset \frac{1}{2}\mathbb{Z}^3$ sitting in the exact sequence

$$0 \rightarrow \mathbb{Z}^3 \mathbf{L} \text{ (windings, } \pi_1) \rightarrow \Gamma \rightarrow \Gamma/\mathbb{Z}^3 \mathbf{L} \text{ (polarization classes)} \rightarrow 0.$$

Ordinary boundary twists realize only the characters of the winding subgroup (flat $U(1)$ connections, $\text{Hom}(\pi_1, U(1))$); the polarization phases of Sec. 11.3 realize the characters of the quotient. Averaging over a subgroup of characters projects onto the annihilator of that subgroup — which is why twist averaging alone keeps half of every winding channel, and why the gauge-invariant label $\delta = \rho + \mathbf{N}\mathbf{L}$ must glue the two factors together.

In plain language

The physical photon is built from local magnetic curls around hexagons. The artifact is a Wilson loop that wraps around the torus but looks short because the torus is small. The transported-dipole character distinguishes these topological classes even when the spin flips look deceptively similar.

11.2 Character-deformed microscopic Hamiltonian

For a nearest-neighbor bond, write the minimum-image displacement as

$$\mathbf{d}_{ij} = \mathbf{r}_j - \mathbf{r}_i - \mathbf{n}_{ij}\mathbf{L},$$

where $\mathbf{n}_{ij}$ records which periodic image was used. The microscopic move $S_i^+ S_j^-$ raises site i and lowers site j . The auxiliary transported-dipole character deformation is

$$H_{\text{mic}}^{2\delta}(\boldsymbol{\theta}) = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} \left[e^{+i\boldsymbol{\theta} \cdot (2\mathbf{d}_{ij})} S_i^+ S_j^- + e^{-i\boldsymbol{\theta} \cdot (2\mathbf{d}_{ij})} S_i^- S_j^+ \right]. \quad (32)$$

This is not an ordinary physical boundary flux. It is a finite-cluster source field conjugate to the integer transported dipole

$$\Delta = 2\delta.$$

For the leading finite-size sectors on the cubic-16 and FCC-32 clusters, $\Delta_\mu = 0, \pm 1$. Thus the minimal grid is

$$\theta_\mu \in \{0, \pi\}, \quad M = 2, \quad (33)$$

with eight character points. If higher-order or larger-cluster diagnostics reveal additional transported-dipole labels, M must be increased until the low-band spectrum and relevant matrix elements converge.

11.3 Structure theorems: the character average is a mask

The deformation (32) looks like it requires a separate microscopic diagonalization at every character point. It does not. Let

$$\hat{X} = \sum_i \mathbf{r}_i S_i^z \quad (34)$$

be the many-body polarization in home-cell coordinates, a *diagonal* operator in the S^z product basis, and define the diagonal unitary $V(\boldsymbol{\theta}) = e^{2i\boldsymbol{\theta} \cdot \hat{X}}$.

Proposition 3 (Isospectrality at the minimal grid). *Conjugating H_{mic} by $V(\boldsymbol{\theta})$ multiplies the hop $S_i^+ S_j^-$ by $e^{2i\boldsymbol{\theta} \cdot (\mathbf{r}_i - \mathbf{r}_j)}$, while the deformation (32) carries $e^{2i\boldsymbol{\theta} \cdot \mathbf{d}_{ij}}$ with $\mathbf{d}_{ij} = \mathbf{r}_j - \mathbf{r}_i - \mathbf{n}_{ij} \mathbf{L}$; the two differ by the boundary factor $e^{-2i\boldsymbol{\theta} \cdot \mathbf{n}_{ij} \mathbf{L}}$. Because $\mathbf{L}$ is an integer matrix, this factor equals 1 at every $\boldsymbol{\theta} \in \{0, \pi\}^3$. Hence*

$$H_{\text{mic}}^{2\delta}(\boldsymbol{\theta}) = V(\boldsymbol{\theta}) H_{\text{mic}} V(\boldsymbol{\theta})^\dagger \quad \text{exactly, at all eight corners.}$$

Consequently all eight corner spectra coincide, and every observable diagonal in the S^z basis — including $P_{\mathcal{I}}$, all Ising correlators, and every thermal trace $\text{Tr } f(H_{\text{mic}}^{2\delta}(\boldsymbol{\theta}))$ — is exactly $\boldsymbol{\theta}$ -independent.

Proposition 3 turns two empirical negative controls into theorems: observable averaging reproduced the bare peak *to all digits* because the character family is isospectral, and the pullback diagnostics were corner-independent because the Gram matrix is.

Proposition 4 (Mask identity). *$P_{\mathcal{I}}$ commutes with $V(\boldsymbol{\theta})$ (both diagonal), so the low-band ice-frame data of Sec. 11 transform covariantly:*

$$X(\boldsymbol{\theta}) = D(\boldsymbol{\theta}) X(0), \quad S(\boldsymbol{\theta}) = S(0), \quad H_{\text{band}}(\boldsymbol{\theta}) = D(\boldsymbol{\theta}) H_{\text{band}}(0) D(\boldsymbol{\theta})^\dagger,$$

with $D(\boldsymbol{\theta})_{\alpha\alpha} = e^{2i\boldsymbol{\theta} \cdot \mathbf{x}_\alpha}$ and $\mathbf{x}_\alpha = \sum_i \mathbf{r}_i S_i^z(\alpha)$ the polarization of ice state α . The eight-corner average is therefore an exact entrywise projection,

$$\boxed{\left(H_{\text{band}}^{\text{clean}} \right)_{\alpha\beta} = (H_{\text{band}}(0))_{\alpha\beta} \left[2(\mathbf{x}_\alpha - \mathbf{x}_\beta) \text{ componentwise even} \right]}, \quad (35)$$

the polarization-parity mask. One untwisted downfolding replaces the eight-point character sweep, and the projection is exact by construction.

Proposition 5 (Factorization of the character group). *For general θ the same algebra gives*

$$H_{\text{mic}}^{2\delta}(\theta) = V(\theta) H_{\text{twist}}(\phi = 2L^T \theta) V(\theta)^\dagger :$$

every character deformation is gauge-equivalent to an ordinary boundary twist composed with a polarization rotation, and the full character average factorizes as

$$(\text{character average over } \hat{\Gamma}) = (\text{ordinary twist average}) \circ (\text{parity mask}).$$

At $M = 2$ the twist factor is trivial and the average is a pure mask; for $M \geq 3$ the twists are genuine and separate diagonalizations are required. This also explains structurally why plain boundary-twist averaging retains exactly half of every winding channel: it implements only the first factor of the product.

All three statements are verified to machine precision against the stored eight-corner data by `notes/verify_mask_identity.py`: corner-spectra spread 3.6×10^{-15} ; zero mask mismatches on all 8100 band-operator entries with residual $< 8 \times 10^{-15}$ at $J_\pm = -0.03$ and -0.05 (1284 of 8100 entries kept); and, as the negative control, the dipole- 4δ , $M = 3$ family — which contains genuine twists — has corner-spectra spread 5×10^{-2} and is *not* reproduced by any mask.

In plain language

The character field looks like eight different Hamiltonians, but at the minimal grid they are all the same Hamiltonian written in eight different phase conventions. All of the cleaning power therefore lives in one place: the fixed ice basis in which the operators are compared. Rotating each copy back into that common frame and averaging simply erases every matrix element whose end states differ by an odd polarization — which is precisely the fingerprint the winding four-loop cannot hide.

11.4 Pullback of the QED low band

Fix one classical ice basis

$$\{|\alpha\rangle\}_{\alpha=1}^{N_{\mathcal{I}}}, \quad P_{\mathcal{I}} = \sum_{\alpha=1}^{N_{\mathcal{I}}} |\alpha\rangle\langle\alpha|.$$

For every character point, QED solves

$$H_{\text{mic}}^{2\delta}(\theta)|\psi_n(\theta)\rangle = E_n(\theta)|\psi_n(\theta)\rangle, \quad n = 1, \dots, N_{\mathcal{I}}, \quad (36)$$

selecting the lowest ice-like band. Define

$$X_{\alpha n}(\theta) = \langle\alpha|\psi_n(\theta)\rangle. \quad (37)$$

The projected vectors $P_{\mathcal{I}}|\psi_n\rangle$ are not exactly orthonormal. Their Gram matrix is

$$S(\theta) = X^\dagger(\theta)X(\theta). \quad (38)$$

The first control diagnostic is that $S(\theta)$ must be nonsingular and well conditioned. Physically, this says the selected microscopic band is still the dressed ice manifold.

Use the Loewdin pullback

$$Q(\theta) = X(\theta)S^{-1/2}(\theta), \quad Q^\dagger(\theta)Q(\theta) = 1. \quad (39)$$

The microscopic low-band operator represented in the fixed ice basis is

$$H_{\text{band}}(\boldsymbol{\theta}) = Q(\boldsymbol{\theta}) \text{diag}(E_1(\boldsymbol{\theta}), \dots, E_{N_{\mathcal{I}}}(\boldsymbol{\theta})) Q^\dagger(\boldsymbol{\theta}). \quad (40)$$

This is not a finite-order Schrieffer–Wolff truncation. It contains the higher-order low-band dressing resolved by the microscopic QED diagonalization at the chosen coupling.

Remark 5 (What the Loewdin pullback means). The word “pullback” is being used in a finite-dimensional Hilbert-space sense. The microscopic eigenvectors $|\psi_n(\boldsymbol{\theta})\rangle$ live in the full Hilbert space, while the gauge Hamiltonian should act in the ice basis. Projecting the microscopic vectors with $P_{\mathcal{I}}$ gives the columns of X :

$$X_{\alpha n} = \langle \alpha | \psi_n \rangle.$$

These projected vectors are generally not orthonormal. Their Gram matrix

$$S_{mn} = \sum_{\alpha} X_{\alpha m}^* X_{\alpha n} = \langle \psi_m | P_{\mathcal{I}} | \psi_n \rangle$$

is the metric induced on the projected low-band frame by the ice-manifold inner product. Multiplication by $S^{-1/2}$ removes this metric distortion:

$$Q^\dagger Q = S^{-1/2} X^\dagger X S^{-1/2} = 1.$$

Thus $Q = X S^{-1/2}$ is the orthonormal frame in the ice Hilbert space that is closest, in the symmetric Loewdin sense, to the projected microscopic low-band vectors.

This is geometric, but not differential geometry in the sense of curvature or differential forms. It is the geometry of finite-dimensional Hilbert-space subspaces: projection, induced metric, and choice of orthonormal frame. The important diagnostic is $\lambda_{\min} S$. If it is small, at least one microscopic low-band direction has little ice weight, the pullback amplifies noise, and the gauge-sector Hamiltonian is not controlled. If S is well conditioned, the QED low band is a dressed version of the ice manifold and Eq. (40) is a meaningful representation of that dressed band as an operator on the ice basis.

11.5 The clean operator

The finite-size-clean gauge-sector Hamiltonian is the trivial transported-dipole character component of Eq. (40). By Proposition 4 it is computed *without any character sweep*:

$$H_{\text{band}}^{\text{clean}} = \text{mask}[H_{\text{band}}(\boldsymbol{\theta} = 0)], \quad \text{mask} : \text{keep } (\alpha, \beta) \iff 2(\mathbf{x}_\alpha - \mathbf{x}_\beta) \in (2\mathbb{Z})^3, \quad (41)$$

which coincides identically with the eight-corner average $\frac{1}{8} \sum_{\boldsymbol{\theta} \in \{0, \pi\}^3} H_{\text{band}}(\boldsymbol{\theta})$. Only after this operator-level projection do we diagonalize. This ordering is essential. A single transverse hop leaves the ice manifold; the local hexagon ring appears only after a complete virtual process returns to the ice manifold. Filtering microscopic hops directly can distort the process we want to keep (the direct microscopic average of Sec. 12 fails for exactly this reason: winding lives in the *composition* of hops — in the π_1 class of the completed path — so no linear operation on H_{mic} can see it). Filtering the downfolded operator acts on the net transported dipole of the completed low-energy process.

In the perturbative limit,

$$H_{\text{band}}^{\text{clean}} - \text{const.} = H_{\text{clean}} + O(J_{\pm}^4 / J_{zz}^3), \quad (42)$$

provided the character grid resolves the transported-dipole sectors present through third order. Away from the strict perturbative limit, Eq. (41) is a numerical gauge-sector downfolding, not a full microscopic all-temperature ED trace.

11.6 Exact implementation checklist (Stage A)

The production implementation is:

1. Build the periodic pyrochlore cluster. Store every nearest-neighbor bond (i, j) , the minimum-image vector $\mathbf{d}_{ij}$, and the periodic image integer $\mathbf{n}_{ij}$.
2. Enumerate tetrahedra and the classical ice basis $\{|\alpha\rangle\}_{\alpha=1}^{N_{\mathcal{I}}}$; record the polarizations $\mathbf{x}_{\alpha} = \sum_i \mathbf{r}_i S_i^z(\alpha)$ *in the same coordinate frame as the cluster builder* and precompute the parity mask of Eq. (35).
3. Diagonalize the lowest $N_{\mathcal{I}}$ states of the *untwisted* H_{mic} using a solver whose eigenvectors are accurately orthonormal. A Krylov or block-Lanczos result is accepted only if its vector-Gram diagnostics pass.
4. Compute X , $S = X^{\dagger}X$, and $Q = XS^{-1/2}$ at $\boldsymbol{\theta} = 0$. Record

$$\lambda_{\min}S, \quad \lambda_{\max}S, \quad \|\Psi^{\dagger}\Psi - 1\|.$$

5. Assemble $H_{\text{band}} = H_{\text{band}}(\boldsymbol{\theta} = 0)$ from Eq. (40) and apply the mask, Eq. (41).
6. Diagonalize $H_{\text{band}}^{\text{clean}}$ and compute

$$C(T) = \frac{\langle E^2 \rangle_T - \langle E \rangle_T^2}{T^2}, \quad S(T) = \ln Z + \frac{\langle E \rangle_T}{T}.$$

7. Compare against the bare low-band operator $H_{\text{band}}^{\text{bare}} = H_{\text{band}}(\boldsymbol{\theta} = 0)$. The bare operator should show the winding four-loop scale, while the clean operator should track g_{hex} in the perturbative window.
8. If double-winding (even nonzero transport) contamination is suspected, refine with genuine boundary twists $\boldsymbol{\phi} = 2\mathbf{L}^T\boldsymbol{\theta}$ on the $M = 4$ grid (Proposition 5) and check stationarity of the clean spectrum.

12 Validation of the QED spine on cubic-16

The 16-site cubic cluster is small enough that every piece of the protocol can be tested against exact microscopic ED. The fixed- $S_{\text{tot}}^z = 0$ Hilbert space has dimension

$$\binom{16}{8} = 12870,$$

and the ice band has $N_{\mathcal{I}} = 90$ states.

12.1 Choosing the character

The perturbative row-table test establishes the minimal character grid. On this cluster, physical flat twists and integer winding projectors do not equal the strict $\boldsymbol{\delta} = 0$ row projector. The $2\boldsymbol{\delta}$ character does:

projector	T_{peak}/J_{zz}	rel. error to $\delta = 0$	comment
all rows	0.0100564	11.6476	winding four-loop present
integer $N = 0$	0.0046462	5.8238	boundary winding only
physical $\mathbf{A} \cdot \delta$, $M = 6$	0.0016563	3.0740	suppresses, not exact
2δ character, $M = 2$	0.0016045	0	minimal row-level charge

This table is a row-level proof of the character. The production calculation is the QED low-band construction in Eq. (41).

12.2 QED low-band results

At $J_{\pm} = -0.05J_{zz}$, the historical QED run used all eight 2δ , $M = 2$ character points; those stored operators are now also the machine-precision verification of Propositions 3 and 4 (identical corner spectra; averaged operator equal to the mask of the $\theta = 0$ operator). The pullback was well conditioned:

$$\min_{\theta} \lambda_{\min} S(\theta) = 0.8872896692,$$

and the saved QED eigenvectors were orthonormal to 2.3×10^{-14} . The specific-heat peaks are

operator	T_{peak}/J_{zz}	comment
QED low band, $\theta = 0$	0.0129476	bare four-loop scale
QED low-band operator average	0.0012227	clean gauge scale
perturbative all rows	0.0100564	four-loop artifact
perturbative $\delta = 0$	0.0016045	leading hexagon target

A second full-QED check at $J_{\pm} = -0.03J_{zz}$ confirms that the clean operator tracks the cubic hexagon scale:

$J_{\pm}/J_{zz}$	$T_{\text{peak}}^{\text{clean}}$	g_{hex}	$T_{\text{peak}}^{\text{clean}}/g_{\text{hex}}$	$T_{\text{peak}}^{\text{bare}}$
-0.03	0.0002996	0.000324	0.925	0.005541
-0.05	0.0012227	0.001500	0.815	0.012948

For the $J_{\pm} = -0.03J_{zz}$ run, the minimum ice-subspace overlap was 0.948 and the full-vector Gram deviation was below 2.1×10^{-14} .

12.3 Rejected shortcuts

The naive direct microscopic average

$$H_{\text{mic}}^{\text{avg}} = \frac{1}{8} \sum_{\theta} H_{\text{mic}}^{2\delta}(\theta)$$

was tested separately. It produces a low band with large ice overlap (0.982 mean at $J_{\pm} = -0.05J_{zz}$), but its specific-heat peak follows g_4 , not g_{hex} :

$J_{\pm}/J_{zz}$	T_{peak}/g_4	$T_{\text{peak}}/g_{\text{hex}}$
-0.02	0.125	2.08
-0.03	0.119	1.32
-0.04	0.114	0.946
-0.05	0.109	0.725
-0.06	0.104	0.581

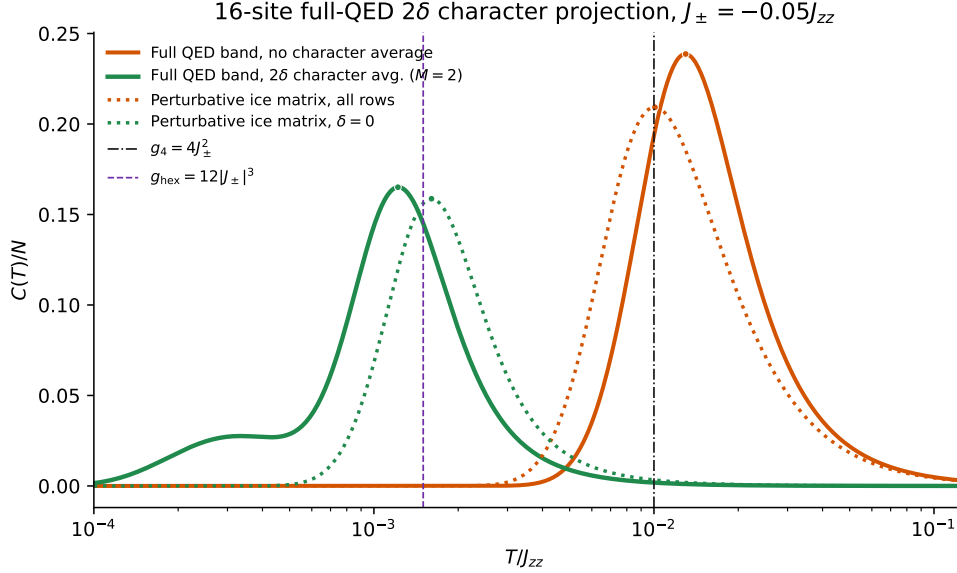


Figure 12: **Specific heat from the adopted QED low-band operator average.** The orange solid curve is the untwisted microscopic low-band operator $H_{\text{band}}(\theta = 0)$. The green solid curve is the operator-level 2δ , $M = 2$ character average. Dotted curves are perturbative ice-manifold matrices. The QED operator average moves the low-temperature peak from the spurious four-loop scale toward the clean hexagon scale.

It changes the coefficient of the finite-size artifact but does not isolate the hexagon manifold.

Observable averaging also fails. At $J_{\pm} = -0.05J_{zz}$, averaging the twist-resolved low-band specific heats gives

$$T_{\text{peak}}/J_{zz} = 0.0129476,$$

the same peak as the bare periodic low band. These shortcuts are useful negative controls, but the spine of the method is Eq. (41).

13 Graded Brillouin–Wigner downfolding: the protocol at any $J_{\pm}$

The Loewdin construction of Sec. 11 anchors the lowest $N_{\mathcal{I}}$ *eigenstates* to the ice basis, and therefore requires the microscopic low band to still *be* an ice band. That requirement fails at material-scale couplings (the pullback Gram matrix becomes singular by $J_{\pm}/J_{zz} \simeq -0.18$ on cubic-16). The all- $J_{\pm}$ engine replaces eigenstate selection by the exact Feshbach–Schur (Brillouin–Wigner) map onto the ice block: with $P = P_{\mathcal{I}}$, $Q = 1 - P$,

$$F(z) = PHP + PHQ(z - QHQ)^{-1}QHP, \quad (43)$$

defined for every $J_{\pm}$ and every $z \notin \text{spec}(QHQ)$. No eigenvectors, no band selection, no Gram matrix. The geometric expansion of the resolvent is a sum over *completed* virtual excursions $P \rightarrow Q \rightarrow \dots \rightarrow Q \rightarrow P$, each carrying a definite transported dipole, so the character grading of Sec. 10 applies to $F(z)$ term by term, resummed to infinite order. Eigenvalues of H with nonzero ice weight are exactly the solutions of the nonlinear problem $\det[F(E) - E] = 0$.

13.1 Equivariance and the masked operator

P and $V(\boldsymbol{\theta})$ are both diagonal in the S^z basis, so the structure theorems transfer verbatim to the BW map:

$$F(z; \boldsymbol{\theta}) = D(\boldsymbol{\theta}) F_{\text{twist}}(z; 2\mathbf{L}^T \boldsymbol{\theta}) D(\boldsymbol{\theta})^\dagger,$$

and at the minimal grid the clean object is a single masked operator,

$$\boxed{F_{\text{clean}}(z) = \text{mask}[F(z; \boldsymbol{\theta} = 0)]}. \quad (44)$$

This projects every virtual process, at every order in $J_\pm$, onto $\Delta = 2\delta = 0 \pmod{2}$: the winding four-loops and wrapping hexagons are removed identically, at any coupling, whether or not perturbation theory converges and whether or not the low band is ice-like.

13.2 Self-consistent solution and built-in diagnostics

The clean gauge-sector levels solve

$$E_n = \text{eig}_n F_{\text{clean}}(E_n), \quad (45)$$

found by a damped fixed-point iteration per level. Three diagnostics come for free:

- **Non-ice weight.** At a solution with ice-block eigenvector u_n , the full eigenstate carries non-ice weight

$$w_n^{\text{out}} = \frac{u_n^\dagger PHQ (E_n - QHQ)^{-2} QHP u_n}{1 + u_n^\dagger PHQ (E_n - QHQ)^{-2} QHP u_n} \in [0, 1),$$

which equals the z -derivative of the eigenvalue of F and is therefore also the convergence rate of the fixed point: the solver converges exactly as fast as the state is ice-like.

- **Pole distance.** $\min_k |E_n - \lambda_k(QHQ)|$ measures the spectral isolation of the gauge-sector resonance from the spinon continuum.
- **Anchoring check.** The *unmasked* self-consistent BW spectrum must reproduce the exact lowest $N_{\mathcal{I}}$ microscopic levels whenever the low band is the ice band; a mismatch is the sharp statement that eigenstate-anchored constructions (Stage A) have lost their target, while the BW resonances remain defined.

Breakdown is therefore *interpretable*: when the solutions collide with $\text{spec}(QHQ)$ and the non-ice weight approaches one, the statement is that the cluster's gauge sector has dissolved into the spinon continuum at that coupling — a physical finding, not a numerical failure.

14 The cubic-16 validation ladder

Every ingredient can be tested exactly on cubic-16 (`notes/masked_bw_16site.py`; records in `notes/masked_bw_16s`). Per coupling the script computes the full microscopic spectrum densely (ground truth), the Stage-A masked Loewdin operator where it exists, and the Stage-B masked BW solution with all diagnostics. Convention anchors: the exact lowest 90 levels agree with the stored eight-corner files to 3×10^{-14} , and the $\boldsymbol{\theta} = 0$ Gram eigenvalue $\eta_{\min}$ reproduces the earlier validity sweep at every coupling.

$J_{\pm}/J_{zz}$	$\eta_{\min}$	conv.	$T_{\text{peak}}^{\text{clean}}/g_{\text{hex}}$	$\max_n w_n^{\text{out}}$	pole gap	Loewdin $/g_{\text{hex}}$	verdict
-0.03	0.948	90/90	0.927	0.029	0.84	0.927	controlled
-0.05	0.887	90/90	0.859	0.074	0.74	0.817	controlled
-0.08	0.790	90/90	0.769	0.194	0.59	0.648	gray zone
-0.10	0.724	90/90	0.705	0.241	0.50	0.540	gray zone
-0.18	0.360	90/90	0.435	0.536	0.185	(0.257)	recovered; strongly dressed
-0.30	~ 0	52/90	—	0.9995	2.1×10^{-4}	—	gauge sector dissolved

Three conclusions, in increasing order of importance:

1. **Reduction.** In the controlled window the masked BW solution coincides with the eight-corner character average. At $J_{\pm} = -0.03$ the two clean specific-heat peaks are identical to all reported digits; the residual maximum spectral deviation (5×10^{-4}) is a uniform shift of the whole band, invisible to thermodynamics. At -0.05 the peaks agree to 5%, and the spread between the two constructions tracks the non-ice weight — the diagnostic doubles as the error bar between resummations.
2. **Extension.** The CHO-like point $J_{\pm}/J_{zz} = -0.18$, where the Gram matrix collapse ($\eta_{\min} = 0.36$) invalidated Stage A, is *recovered*. The anchoring check makes the mechanism explicit: the unmasked BW resonances differ from the exact lowest-90 spectrum by up to $4.2 J_{zz}$ there (five levels do not even converge to the lowest band), so the lowest eigenstates are simply no longer the ice band; but the ice-block resonances themselves still exist, all 90 masked levels converge, and they remain separated from the spinon continuum by $0.185 J_{zz}$. The old “failure” was an artifact of eigenstate anchoring, not of the physics.
3. **Honest boundary.** At the CZO-like point $J_{\pm}/J_{zz} = -0.30$ the diagnostics deliver a verdict instead of a crash: 38 of 90 levels do not converge, the surviving solutions sit $2 \times 10^{-4} J_{zz}$ from the continuum poles, and the non-ice weight reaches 0.9995. On this cluster, at this coupling, there is no isolated gauge sector to clean. Whether FCC-32 restores one is now a well-posed question for the same diagnostics.

The status classification is now three-tiered and applies at any coupling: *controlled* ($w^{\text{out}} \lesssim 0.1$, pole gap $O(J_{zz})$: production gauge-sector ED), *gray zone / exploratory* ($0.1 \lesssim w^{\text{out}} \lesssim 0.5$: report with the BW–Loewdin spread as the method error), and *dissolved* (unconverged levels, pole gap $\rightarrow 0$, $w^{\text{out}} \rightarrow 1$: the finding is the absence of an isolated gauge sector, and only a larger cluster can change it). The sign of $J_{\pm}$ plays no role in the classification; the zero-flux side additionally admits sign-free QMC benchmarks.

15 All-temperature consistency check on cubic-16

The QED low-band operator average is a gauge-sector construction. By itself it does not contain the larger-temperature peak produced by sectors with violated tetrahedron constraints. Therefore the all-temperature diagnostic must alter only the part of the spectrum where the finite-size pathology lives.

Let

$$E_1^{\text{full}} \leq E_2^{\text{full}} \leq \dots \leq E_{2^{16}}^{\text{full}}$$

be the exact untwisted microscopic spectrum, including all S_{tot}^z sectors, and let

$$\epsilon_1^{\text{clean}} \leq \dots \leq \epsilon_{N_{\text{ice}}}^{\text{clean}}$$

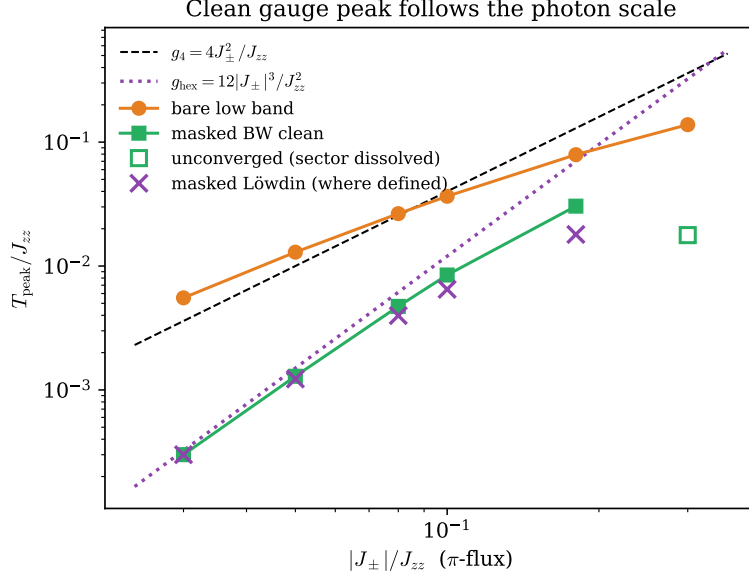


Figure 13: **Peak scaling across the full coupling range (cubic-16).** The bare low band (orange) tracks the winding artifact g_4 ; the masked BW clean operator (green) tracks the photon scale g_{hex} ; crosses are the masked Loewdin operator where its Gram matrix is nonsingular.

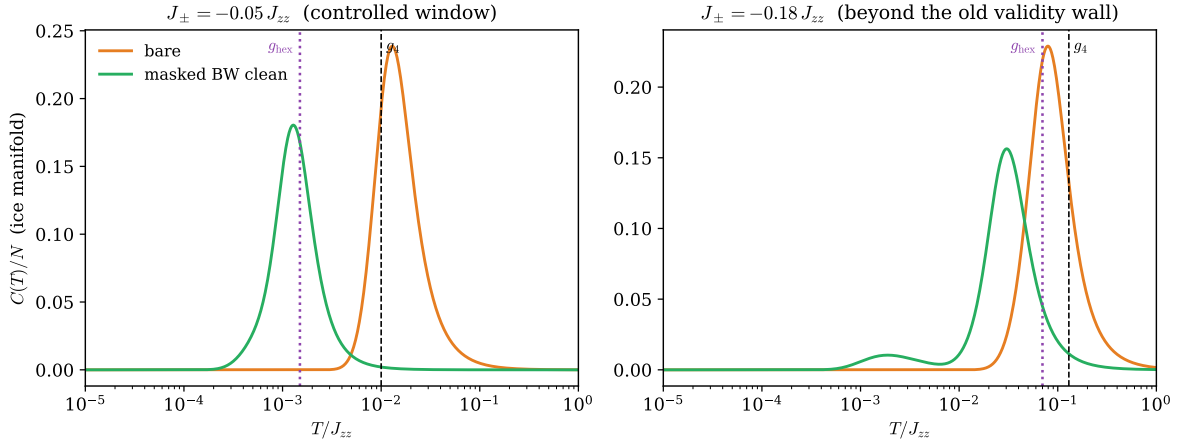


Figure 14: **Clean specific heat in and beyond the old validity window.** Left: the controlled point $J_{\pm} = -0.05 J_{zz}$. Right: the CHO-like point $J_{\pm} = -0.18 J_{zz}$, inaccessible to the eigenstate-anchored protocol, where the masked BW operator still yields fully converged clean thermodynamics (reported with its 54% maximum dressing diagnostic). Dashed and dotted verticals mark g_4 and g_{hex} .

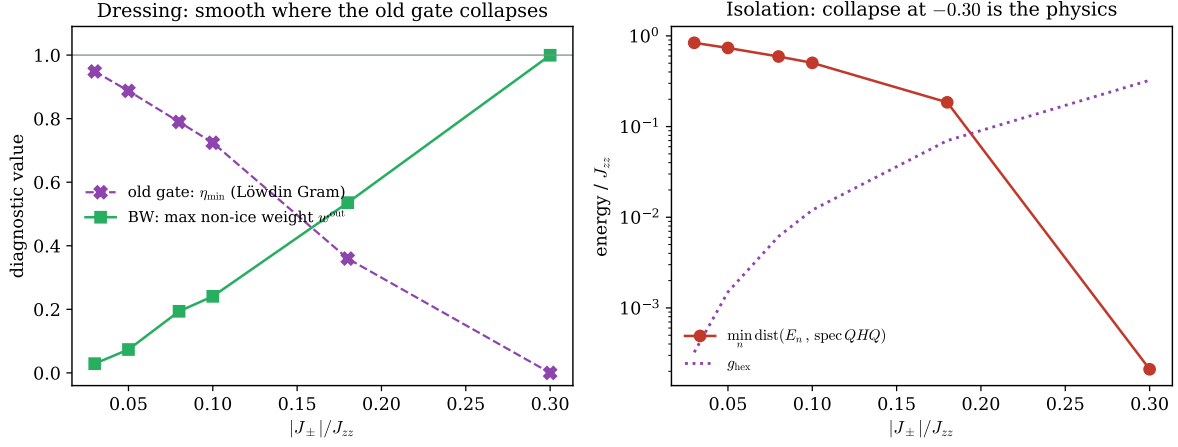


Figure 15: **All- $J_{\pm}$ validity diagnostics.** Left: the old Stage-A gate $\eta_{\min}$ (purple) against the BW non-ice weight (green); the former collapses at -0.18 while the latter degrades smoothly and only saturates at -0.30 . Right: minimum distance of the solved levels to $\text{spec}(QHQ)$ — the spectral-isolation diagnostic whose collapse at -0.30 constitutes the physical statement that the gauge sector has merged with the spinon continuum.

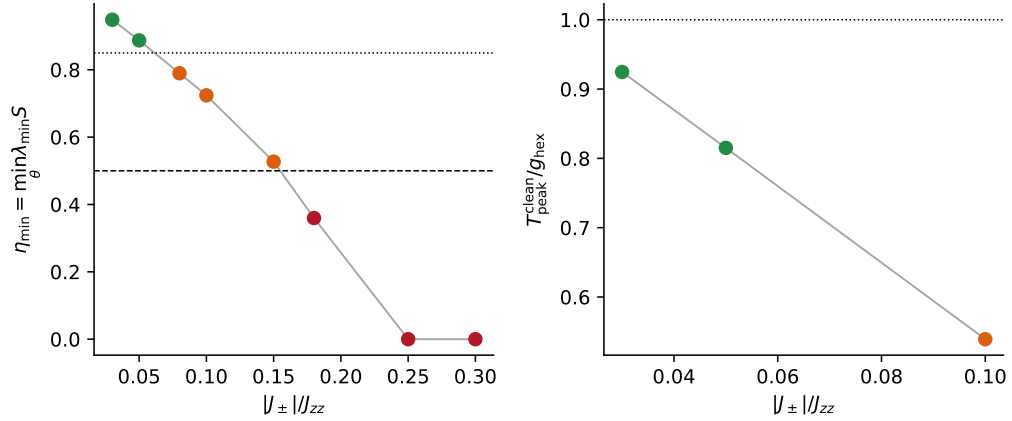


Figure 16: **Historical Stage-A validity sweep** (Löwdin pullback $\eta_{\min}$ and eight-corner clean peaks), retained for comparison with the BW ladder above.

be the spectrum of $H_{\text{band}}^{\text{clean}}$. Define

$$\tilde{E}_a = \begin{cases} \epsilon_a^{\text{clean}} + s, & 1 \leq a \leq N_{\text{ice}}, \\ E_a^{\text{full}}, & a > N_{\text{ice}}, \end{cases} \quad s = \frac{1}{N_{\text{ice}}} \sum_{a=1}^{N_{\text{ice}}} E_a^{\text{full}} - \frac{1}{N_{\text{ice}}} \sum_{a=1}^{N_{\text{ice}}} \epsilon_a^{\text{clean}}. \quad (46)$$

The shift s preserves the center of the ice band relative to the spinon continuum and changes only the internal gauge-sector splittings.

QED exact full-spectrum ED gives all $2^{16} = 65536$ eigenvalues. The lowest 90 exact eigenvalues agree with the untwisted low-band extraction to 6.22×10^{-14} . The peak comparison is

curve	low- T peak	larger- T peak
exact full ED	0.0129475	0.252820
full ED with clean ice-band replacement	0.0012227	0.252931

The larger peak shifts by only 4.37×10^{-4} relatively, with height change -4.83×10^{-3} . The finite-size correction acts where intended: on the low-energy ice-band winding artifact, not on the microscopic constraint-violation scale.

16 FCC-32 production protocol

The $2 \times 2 \times 2$ FCC pyrochlore cluster has 32 spins and

$$N_{\mathcal{I}} = 2970$$

ice states. Full microscopic ED of all 2^{32} states is unavailable, so the production calculation is the gauge-sector construction above. By Proposition 4 it requires a *single* untwisted low-band extraction:

$$H_{\text{band,FCC32}}^{\text{clean}} = \text{mask}[H_{\text{band,FCC32}}(\boldsymbol{\theta} = 0)], \quad H_{\text{band,FCC32}}^{\text{bare}} = H_{\text{band,FCC32}}(\boldsymbol{\theta} = 0),$$

an eightfold cost reduction relative to the character sweep, with the projection exact rather than grid-sampled. At material-scale couplings the Stage-B object replaces the band extraction: 2970 shifted block-Krylov solves $(z - QHQ)^{-1}QHP e_\alpha$ in the fixed- S^z sector (dimension 601,080,390), one Krylov space per right-hand side serving all z , plus a Lanczos estimate of the QHQ edge for the pole-gap diagnostic.

The FCC implementation must:

1. express $\mathbf{d}_{ij}$, $\boldsymbol{\Delta} = 2\boldsymbol{\delta}$, and the polarizations $\mathbf{x}_\alpha$ in the FCC primitive coordinate frame used by the cluster builder, and unit-test the parity mask against the cubic-16 reference (`notes/verify_mask_identity.py` logic);
2. run the untwisted low-band extraction (Stage A) or the BW solves (Stage B) with ice-weight, Gram, pole-gap, and non-ice-weight diagnostics;
3. diagonalize both $H_{\text{band,FCC32}}^{\text{bare}}$ and $H_{\text{band,FCC32}}^{\text{clean}}$;
4. check that the clean peak scales with g_{hex} at small $|J_\pm/J_{zz}|$;
5. spot-check M -convergence with genuine boundary twists $\boldsymbol{\phi} = 2\mathbf{L}^T\boldsymbol{\theta}$ on the $M = 4$ grid (Proposition 5);

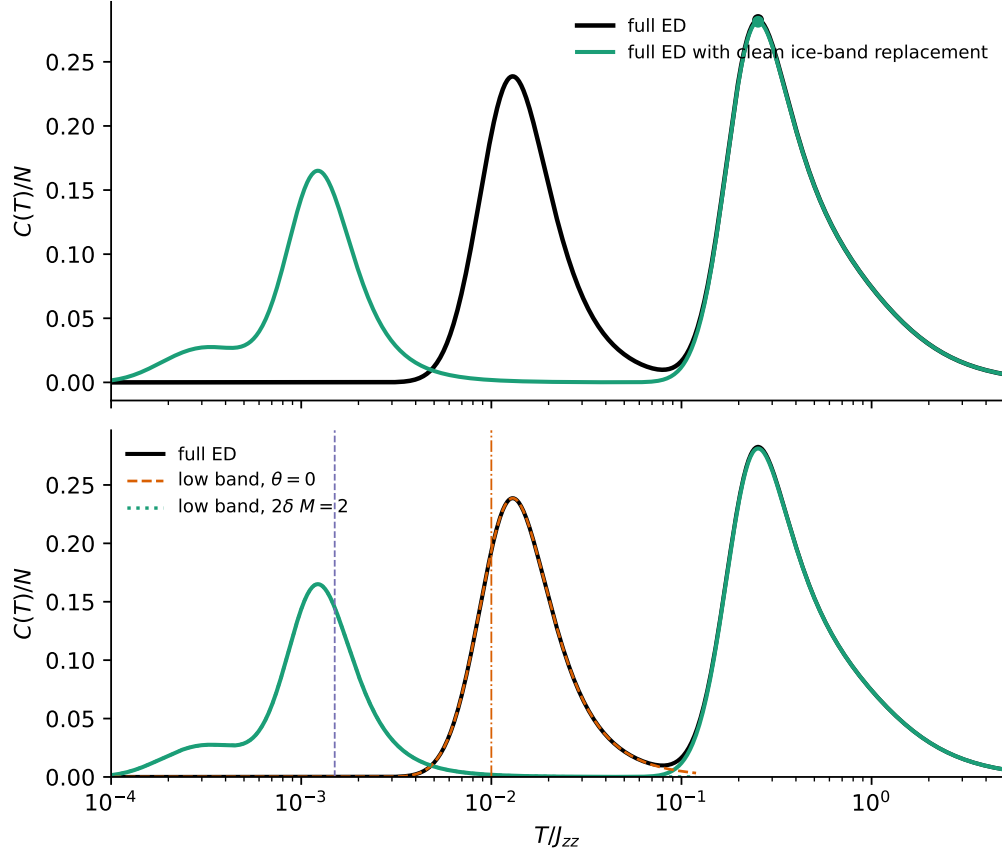


Figure 17: **All-temperature full-ED consistency check.** The black curve is the exact untwisted microscopic full-ED specific heat for all 65536 states of the 16-site cluster. The green curve keeps every non-ice eigenvalue fixed and replaces only the lowest 90 ice-band eigenvalues by the clean QED low-band operator spectrum, with the band-center shift in Eq. (46). The correction moves the spurious low-temperature four-loop peak while leaving the larger full-ED peak at $T/J_{zz} \simeq 0.253$ in place.

6. benchmark the zero-flux side against sign-free QMC before making pi-flux material claims.

The material-facing runs should be interpreted as gauge-sector predictions. For the first U(1)-symmetric campaign one sets $J_{\pm\pm} = 0$ after rotating away J_{xz} and uses representative pi-flux points,

$$\text{CZO-like: } J_{zz} = 1, \quad J_{\pm} = -0.30, \quad J_{\pm\pm} = 0,$$

$$\text{CHO-like: } J_{zz} = 1, \quad J_{\pm} = -0.18, \quad J_{\pm\pm} = 0.$$

These are not zero-flux material partners. Zero-flux calculations are generic sign-free QMC benchmarks for the method.